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SECP3623 ENTERPRISE SYSTEM DESIGN AND MODELING

(WBL)

SECTION 02

PROJECT:

SUBJECT REGISTRATION & ACADEMIC ADVISING

LECTURER NAME: DR NOORMINSHAH BINTI A.IAHAD

GROUP MEMBERS:

NAME	MATRIC NUMBER
BRENDAN CHIA YAN FEI	A23CS0211
GUI KAH SIN	A23CS0080
NEO LI XIN	A23CS0253
SABRINA HENG WEI QI	A23CS0265
WOO CHENG SHUAN	A23CS0283

CHAPTER 1

INTRODUCTION

1.1 Problem Background

Universiti Tun Hussein Onn Malaysia (UTHM) is a public technical university that places strong emphasis on engineering, technology and digital transformation in education. To support academic and administrative procedures, UTHM having systems like Student Information System (SMP), Staff Information System (SMS), Single Sign-On (SSO) portal and the My UTHM mobile application.

Subject registration is one of the most critical academic processes in higher education and it has a direct impact on students' academic planning, timetable management and graduation process. Although UTHM's existing Student Information System provides basic subject registration functionality, the process still faces limitations in terms of usability, validation and work efficiency.

In this project, our group proposes the Subject Registration Enhancement Module, a subsystem that used to enhance the current subject registration process by having new features such as structured course browsing, timetable conflict checker, credit hour validation and a digital add or drop application and approval workflow. The proposed module is designed based on Enterprise Architecture (EA) principles to ensure integration, scalability and alignment with UTHM's existing digital ecosystem.

1.2 Problem Statement

A smooth academic experience for both staff and students is hampered by a number of issues with Universiti Tun Hussein Onn Malaysia's (UTHM) present digital ecosystem. In particular, the following problems have been found:

- Data silos and system fragmentation: Although there are systems like MyUTHM and the Student Information System (SMP), they frequently operate independently. This results in data duplication, as faculty records and student information are stored independently across many apps, leading to information inconsistencies.
- Prevalence of Manual Processes: Manual interventions, paper-based forms, or disjointed technologies like Google Forms and Excel continue to play a major role in critical academic workflows like add/drop requests and registration modifications. This slows down the approval process and adds to the administrative load.
- Absence of Real-Time Validation and Analytics: At the time of selection, the current registration interface does not provide real-time integrated functionality to identify schedule conflicts or validate credit hour restrictions. Without these automated checks, students commonly experience scheduling mistakes that require human rectification later.
- Limited Mobile Integration: Despite the availability of the MyUTHM mobile app, a number of students are facing procedures — particularly course registration and add/drop workflows — that remain unintegrated or necessitate access through several online portals, limiting a cohesive student experience.

1.3 Project Objective

- Enhance registration accuracy by implementing real-time validation logic that automatically detects timetable conflicts and enforces credit hour limits (min/max) prior to submission.

- Streamline the Add/Drop workflow to reduce manual processing time by digitizing the application and advisor approval process, eliminating the need for paper-based or disjointed communication.
- Centralize course data availability by developing a searchable Course Catalogue Browser that integrates directly with UTHM's existing SMP master data to ensure students access up-to-date programme information.

1.4 Project Scope

1. Course Registration & Application Portal

   1.1 Display a list of courses based on student's registered programme

   1.2 Search courses through specific keywords

   1.3 Filter courses by vacancy status, timetable compatibility and credit hours limit

2. Timetable Conflict Checker

   2.1 Cross-reference selected courses against existing registered courses

   2.2 Detect time conflicts between pending selected courses

   2.3 Highlight conflicted and duplicated subject selections

3. Credit Hour Validator

   3.1 Calculate total registered credit hours for the semester

   3.2 Validate total credit hours for each programme's minimum and maximum limits

   3.3 Automatically show warning alerts if credit hours exceed or insufficient

4. Add/Drop Approval Workflow

   4.1 Allow advisor to view and filter requests by students

4.2 Approve or reject applications with optional comments

4.3 Notify student through automated email

5. Course Master Data Administrator

   5.1 Create, update and archive course details

   5.2 Configure number of sections per course subject and capacity quotas

   5.3 Set registration rules and prerequisites

1.5 Project Importance

The importance of the Subject Registration Enhancement Module is that it strengthens one of the most critical academic processes at UTHM which are course registration and academic planning. At the university level, the project supports UTHM’s digital transformation strategy by reducing dependence on fragmented systems, manual verification and approval processes, as well as repetitive administrative work. By embedding real-time validation rules, the system improves data accuracy, reduces error correction tasks, and ensures academic schedules such as timetable compatibility and credit hour limits are consistently enforced.

For academic departments and faculty advisors, the module introduces a structured Add/Drop approval workflow that replaces informal communication channels and paper-based procedures. This ensures that every decision is recorded, traceable, and aligned with programme requirements, thereby improving governance and accountability.

For the student perspective, the project improves usability, efficiency, and clarity in subject registration. Features such as a searchable course catalogue, timetable conflict detection, and automated merit calculation help students make informed choices, avoid registration mistakes, and plan their academic easily and strategically. This reduces stress, minimizes registration delays, simplifies the preparation process, and supports smoother academic progression.

Overall, the project is significant because it benefits the organization through process efficiency and better data management, while simultaneously enhancing the user experience for students, academic advisors, and administrators. It demonstrates how Enterprise Architecture principles can guide subsystem design that integrates effectively with existing system such as SMP, SMS, and SSO, ensuring sustainability and scalability in the long term.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

At UTHM and similar institutions, digital transformation isn't a buzzword; it's a necessity. The only way forward is a structured approach to building systems that actually work together, and Enterprise Architecture (EA) sits right at the heart of this. EA is not just a theory; it's the map that ties big-picture goals to the real nuts and bolts of technology.

This chapter evaluates the specific functional requirements of course registration, such as catching timetable clashes as they happen and making sure students don't sign up for too many credits. There's also a deep dive into why the team picked SAP Build Apps for this project, and how it fits with tried-and-true development methods like Agile or the Software Development Life Cycle (SDLC). To build a modern, flexible prototype that actually works at scale, not just in theory. In the end, this review connects the dots between big architectural ideas and the real-world job of building a better subject registration system.

2.2 Enterprise Architecture (EA)

Enterprise Architecture (EA) is a structured approach used to align an organization's business processes, information systems, data and technology infrastructure with its strategic objectives. EA provides a holistic view of an organization by defining how different layers—business, application, data, and technology—interact with each other.

EA plays an important role in academic and administrative management in higher education institutions. Universities frequently run a variety of information systems for various purposes, including learning management, staff administration, student management and financial management. This system may function in silos without a clear architectural framework, which would result in redundant data and inefficiencies.

Previous studies on University Enterprise Architecture highlight that EA improves system integration, supports digital transformation, and enhances decision-making by providing standardized processes and shared data models. Applying EA principles ensures that newly developed subsystems, such as subject registration modules, can be integrated seamlessly into existing enterprise systems.

2.3 Related Sub-System: Subject Registration Enhancement Module

2.3.1 Analysis of Previous and Existing UTHM Systems

Subject registration at UTHM has evolved through several generations of digital platforms, each aiming to reduce manual administrative burdens:

- **SMAP Online (Sistem Maklumat Akademik Pelajar):** This is the core module within the broader Sistem Maklumat Pelajar (SMP). It handles basic course enrollment, credit exemption, and performance tracking. However, students have reported technical issues such as data loss when switching devices and a lack of real-time conflict detection during the registration for popular subjects.
- **CENTRAL Portal (Core Enterprise Technology Portal):** Launched in November 2021, this framework is UTHM's current initiative to bring separate systems together into a unified, mobile-responsive portal. While it integrates SMP and SMS (Staff Information System), the core registration logic still functions as a digitised version of older processes rather than an enhanced, intelligent system.
- **MyUTHM Mobile Application:** A mobile-centric interface providing virtual IDs and links to SMAP. However, its current registration functionality is often

limited to viewing information rather than executing complex, validated workflows like digital Add/Drop approvals.

2.3.2 Limitations in Existing Systems

The Subject Registration Enhancement Module fixes several critical gaps still found in the CENTRAL portal.

1. **Asynchronous Advising Workflow:** Current Academic Advising is often managed via separate modules or offline consultations. This leads to information inconsistencies and redundant data storage where an advisor's approval is not immediately reflected in the student's registration status.

2. **No Real-Time Verification:** SMAP lacks integrated, real-time logic. Students only discover timetable clashes or credit hour violations after submission attempts, leading to a trial-and-error experience.

3. **Repetitive Manual Course Changes:** Many faculty-level changes still require students to fill out physical forms or Google Forms which are then manually entered into the system by administrative staff.

2.3.3 Proposed Enhancements

By utilising Figma to develop high-fidelity prototypes, this enhancement adds a visual and functional processing tier to the current SMP Master Data. The module implements a Timetable Conflict Checker and Credit Hour Validator to assist students during selection. This approach aligns with Enterprise Architecture (EA) by ensuring that registration records, advising status, and staff approvals remain consistent across both the business and data layers of the UTHM ecosystem.

2.4 Technology Used

In order to meet the project goals of the fast development and the smooth integration with the external Enterprise Architecture of UTHM, the main technology to be used in the development of the Subject Registration Enhancement Module is SAP Build Apps.

SAP Build Apps is an enterprise low-code application development platform, which allows the creation of pixel-perfect web and mobile apps, using the same codebase. Three strategic considerations that were within the scope of the project prompted the selection of this technology:

1. Rapid Prototyping and Logic Implementation: The visual logic flow of the platform enables the direct implementation of complex validation rules, e.g., the Timetable Conflict Checker and Credit Hour Validator, without the heavy amount of coding. This makes sure that it is based on architectural logic and business process optimization as opposed to syntax debugging.

2. API Integration Capabilities: To resolve the problem statement issue of Data Silos, SAP Build Apps has the ability to support strong REST API integration. This enables the new module to conceptually read and write data to the legacy systems (SMP and SMS) of UTHM in real-time to simulate a unified data layer where student records and course quotas are changed in real time across the ecosystem.

3. Cross-Platform Accessibility: This technology will help deal with the problem of "Limited Mobile Integration" whereby the module will be responsive by default. It enables the application that is developed as a result to be rolled out as a web portal to be used by the administration and a mobile-friendly application to be used by students, all of which make the user experience similar despite the current digital landscape.

2.5 Summary

Universities often face challenges from fragmented digital systems, duplicated data, and semi-manual workflows, particularly in subject registration and academic advising. At UTHM, existing systems such as SMP, SMS, SSO, and the MyUTHM mobile application provide basic functionalities but lack real-time validation, automated approvals, and integrated workflows. These limitations result in timetable conflicts, credit hour violations, repeated manual corrections, and inefficiencies for the students, academic advisor, and administrative staff.

Enterprise Architecture (EA) provides a structured framework to address these issues by ensuring that new subsystems are designed to integrate seamlessly with the existing platforms through shared data structures, standardized processes, and consistent workflow. The traditional registration systems often offer only basic enrollment functions and rely heavily on semi-digital processes, particularly for Add/Drop requests and advising approvals. Modern digital initiatives, including low-code platforms and API-based integration, further enable rapid development and interoperability while maintaining alignment with institutional systems.

The proposed Subject Registration Enhancement Module leverages these principles to improve subject enrollment and academic advising. The key features include a searchable course catalogue, timetable conflict checker, credit hour validator, digitized Add/Drop approval, and decline workflow. By integrating with the core systems such as SMP, SMS, and SSO, the module enhances data accuracy, reduces manual processing, streamlines administrative tasks, and provides a cohesive and efficient user experience. This approach demonstrates the practical application of EA concepts and digital transformation strategies within the UTHM’s ecosystem.

CHAPTER 3

METHODOLOGY

3.1 Introduction

The purpose of this chapter is to provide a comprehensive overview of the methodology and technical framework employed to develop the Subject Registration Enhancement Module for Universiti Tun Hussein Onn Malaysia (UTHM). It outlines the systematic steps taken to ensure that the project not only adheres to established Enterprise Architecture (EA) principles but also effectively addresses the specific functional gaps identified in the current registration system.

For this project, a prototyping approach within the Software Development Life Cycle (SDLC) has been selected as the primary development framework. This choice is strategically motivated by the use of Figma as the primary design tool to create high-fidelity, interactive prototypes. By utilising Figma, the project team can focus on user experience (UX) and interface (UI) optimization, ensuring that features are intuitively designed before full-scale development. This methodology ensures that the proposed subsystem remains user-centric and aligns seamlessly with UTHM's broader digital ecosystem, including the Student Information System (SMP) and Staff Information System (SMS).

3.2 Chosen Methodology

The Waterfall methodology was used by our group due to its provision of a well-documented, sequential, and structured development process which is appropriate in academic projects with known requirements. Waterfall is the opposite of Agile as it does not rely on the regular feedback with the company. This is because we cannot afford to hold periodical meetings or reviews of the requirements with UTHM, Waterfall enables us to compile the requirements at the initial stages, complete the design and continue with the tasks step wise without necessarily having to involve the management and other staff of the university on a regular basis.

Waterfall was also selected since our subsystem involves well-spread academic processes like course browsing, timetable conflict check, credit hour validation and advisor approval logic. All these processes can be fully modelled internally and validated by using Figma prototyping and documentation and not have to be live iterated with UTHM. The methodology helps our group to accomplish every stage of the process requirement analysis, UI/UX design, system modelling, and documentation and proceed to the implementation planning stage with minimal disruption and responsibility of groups members. Thus, Waterfall approach is the most realistic and manageable one when it comes to the design of an academic sub-system that can conceptually fit in the digital ecosystem of UTHM and still address the requirements of project submission in an efficient way.

3.3 Phases of Methodology

In contrast to the full six-step Waterfall methodology, our subsystem will cover only the first three phases that directly support our project development.

i. Requirements Analysis

The first phase involved understanding the existing subject registration process and identifying system requirements. This phase focused on reviewing current workflows within UTHM, analysing each users' roles, and identifying existing issues related to timetable clashes, manual approvals and data inconsistencies.

Key activities:

- Reviewing existing features in UTHM academic and administrative systems
- Defining user roles (student, academic advisor and administrator)
- Developing detailed functional requirements and constraints

ii. **System and Architecture Design**

In this phase, the overall system structure was modelled according to Enterprise Architecture principles to ensure alignment with UTHM's digital ecosystem. The design focused on clarifying how the subsystem would conceptually integrate with the existing system as well as the other submodules while supporting consistent data flow and smooth process.

Key designs:

- Enterprise Architecture
- System Architecture Diagram
- Conceptual Data Model (ERD Diagram)
- Use Case Diagram and Description
- Activity and Sequence Diagram

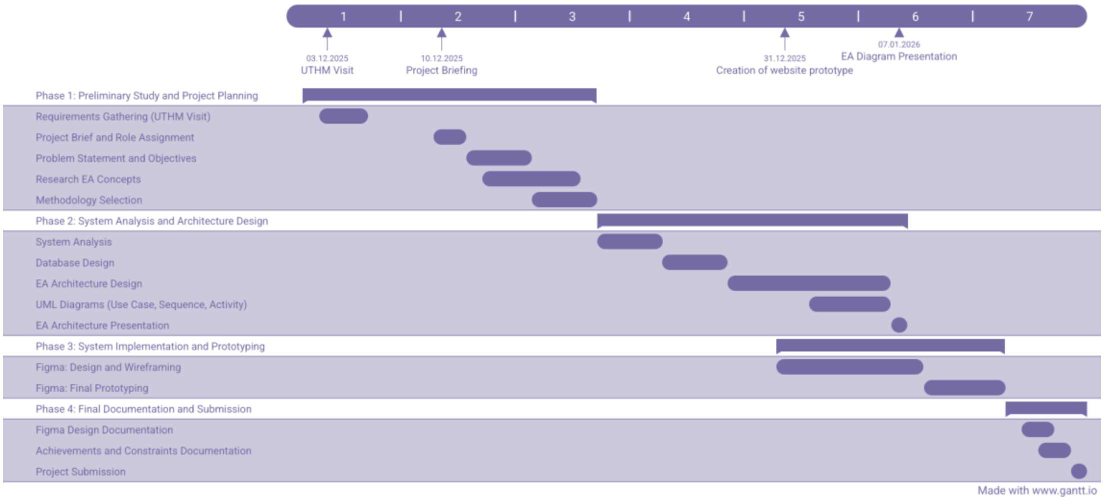
iii. **User Interface and Prototype Development**

The design concepts were then transformed into high-fidelity interactive prototypes using Figma. Rather than full system coding, the prototype was developed to simulate user interactions and system responses.

Key features:

- Visualizing page layouts and navigation flow
- Demonstrating conflict detection and validation behaviour
- Ensuring logical consistency across interfaces and processes before the real implementation was undertaken.

3.4 Project Planning Schedule



Phase 1: Preliminary Study and Project Planning (Dec 03, 2025 – Dec 19, 2025)

The project was started formally with a preliminary feasibility study on December 3, 2025, where we carried out the early requirement gathering interviews to confirm the problem statement before the official start of the project on December 10, 2025. In this preliminary stage, we determined the scope of the project, objectives, and importance as stated in Chapter 1. At the same time, an in-depth Literature Review (Chapter 2) was performed to study the Enterprise Architecture (EA) concepts and examine the available subsystems. This stage ended with the choice of the development methodology and the determination of project schedule in Chapter 3.

Phase 2: System Analysis and Architecture Design (Dec 20, 2025 – Jan 07, 2026)

During this phase, we discussed the weaknesses of the existing system and developed a new Enterprise Architecture solution, which included a description of the Business, Application, Data, and Technology layers. Additionally, the Use Case, Activity and Sequence diagram are designed for each submodule of the Subject Registration and Academic Advising enhancement module. Later on January 7, 2026, we presented our Enterprise Architecture (EA) diagram of our enhancement module to the UTHM representative. This presentation acted as a validation gate way, which made the architectural design fit the strategic objective of the organization before the final prototype was created.

Phase 3: System Implementation and Prototyping (Dec 31, 2025 – Jan 13, 2026)

Concurrently with the last design phase, the implementation phase was initiated on December 31, 2025, and the development of a High-Fidelity prototype started with the use of Figma. This stage was characterized by the translation of the person interface designs into an interactive user experience to emulate the logic of the system requirements (Input, Process, and Output).

Phase 4: Final Documentation and Submission (Jan 14, 2026 – Jan 18, 2026)

After the Figma final testing, we will concentrate on completion of the documentation, especially Chapter 5: Conclusion that will entail the contributions, constraints and recommendations that the system made to the future work. The remaining time will be spent on drafting the complete report, ensuring standard document format and creating softcopy and hardcopy summaries to be submitted to the end deadline of January 18, 2026.

3.5 Summary

This chapter introduced the approach that was used in the creation of the Subject Registration Enhancement Module in Universiti Tun Hussein Onn Malaysia (UTHM). A formal Waterfall methodology as part of the Software Development Life Cycle (SDLC) was chosen because of its applicability to academic projects that have clear requirements and have little stakeholder engagement. The methodology was used to make sure that system analysis, architectural design, and prototype development were done in a logical and systematic way.

The project was implemented in distinct stages, where the first stage was requirements analysis, then system and architecture design and finally high-fidelity prototyping with Figma. An elaborate project schedule was also drawn to facilitate the development process and to make sure that it is completed on time. In general, the given methodology offered a rigorous framework that facilitated the implementation of Enterprise Architecture principles and make sure that the suggested subsystem would be compatible with the current digital ecosystem and academic processes of UTHM.

CHAPTER 4

ANALYSIS AND DESIGN

4.1 Introduction

This chapter presents the analysis and design of the proposed Subject Registration Enhancement Module for Universiti Tun Hussein Onn Malaysia (UTHM). It is concerned with the current process of subject registration and pinpoints its weaknesses and suggests a better designed system that can resolve the problems like conflict of timetables, credit hour validation and ineffective add or drop procedures. The analysis will be done based on the UTHM case study and aims to enhance academic administration through a more integrated and user-friendly digital solution.

The chapter further outlines the system requirements and applies Enterprise Architecture principles to ensure alignment between business processes, application components, data structures and supporting technologies. Design artefacts such as use case diagrams, activity diagrams, sequence diagrams, database designs and interface mock-ups are used to illustrate the proposed system clearly. Overall, this chapter serves as a blueprint for the implementation of the Subject Registration Enhancement Module and supports UTHM's vision toward a Smart Campus.

4.2 Company Organization Structure

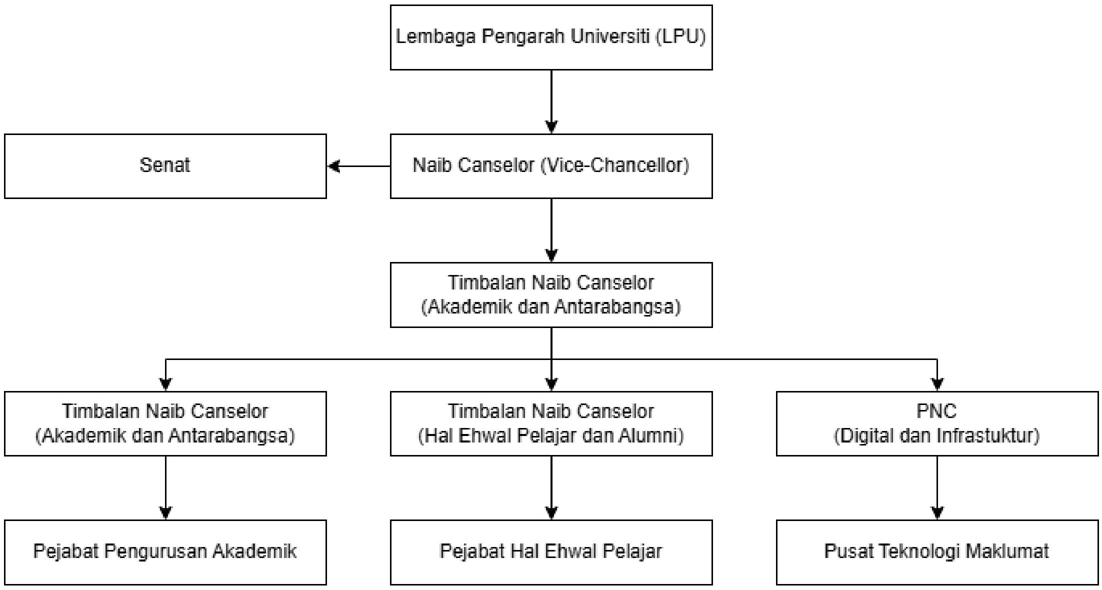


Figure 4.1 UTHM Organization Structure and Subsystem Stakeholder

In Figure 4.1, there is the organisational structure of Universiti Tun Hussein Onn Malaysia (UTHM) in relation to academic governance and digital academic processes. The institutional level begins with Lembaga Pengarah Universiti (LPU) which is the highest governing body that establishes policy and strategic direction of the institution. Below it is the Naib Canselor (Vice-Chancellor) who is in charge of the overall university administration. Timbalan Naib Canselor (Academic and International) subsequently controls academic administration and international academic alignment and thus this position is directly linked with subject registration planning and academic advising decisions.

On the operational level, the structure divides to three important areas. The Pejabat Pengurusan Akademik organizes programme and course management, the courses and subjects being offered and registration policies. The Hal Ehwal Pelajar Pejabat is student management and student support services conceptually related to advising records. The Pusat Teknologi Maklumat is in charge of digital infrastructure and authentication, such as systems such as Student Information System (SMP), Staff Information System (SMS), and MyUTHM. The chart is significant to our project since it defines stakeholders in the ownership of academic data, in the approvals, and the support of the system which will assist us to place our subsystem in an objective position within the institutional environment and the flow of governance of UTHM.

4.3 Current System Analysis

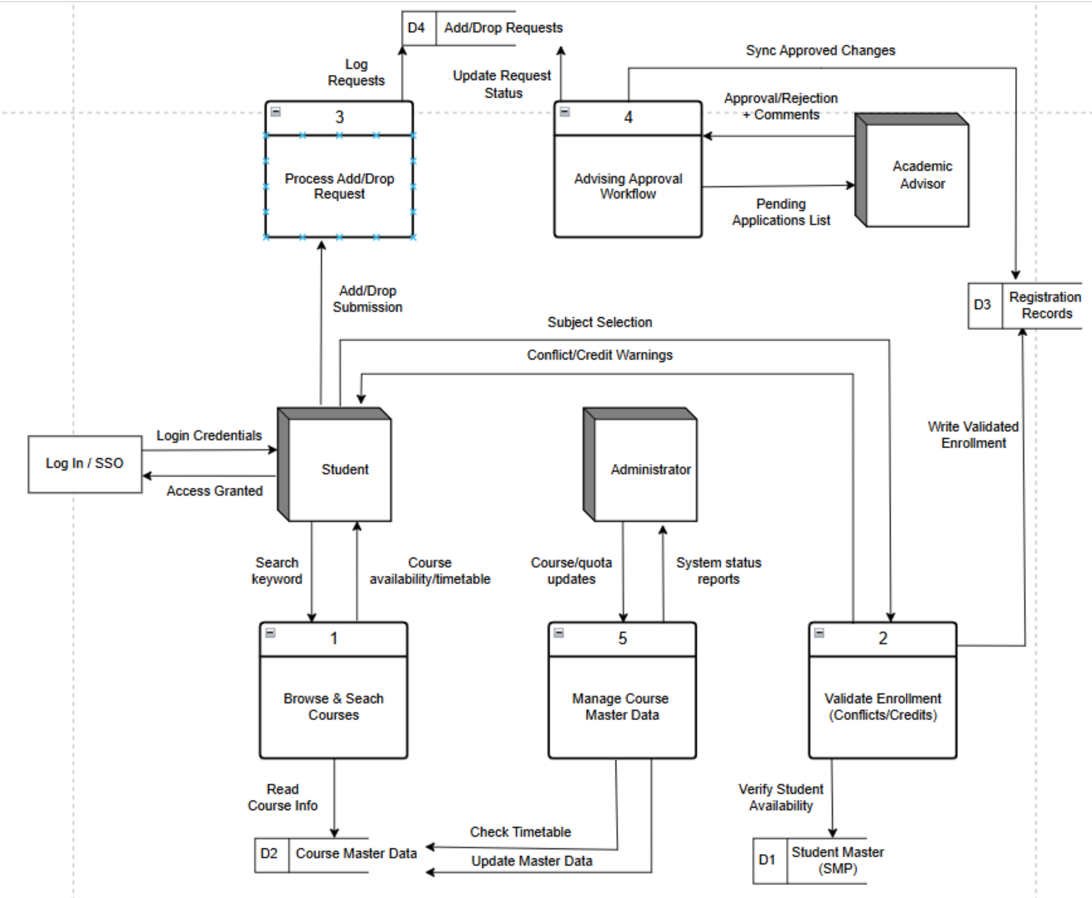


Figure 4.2 Current System Diagram

The subject registration process at Universiti Tun Hussein Onn Malaysia (UTHM) is affected by poor data integration between the Student Information System (SMP) and related stakeholders. In this manual-heavy environment, students initiate the process by submitting course codes through the SMAP Online portal, yet critical modifications such as add or drop requests still rely on separate, paper-based forms or external digital tools like Google Forms.

The administrative and advisory components are also affected by poor system integration. Academic advisors review registration requests through informal channels such as email or physical documents due to the absence of a centralized approval dashboard. Consequently, administrative staff must manually update faculty-level changes in the SMP, increasing workload and the risk of human error. Although advisors can access student records via the Staff Information System (SMS), these records often lack real-time updates from the registration portal, resulting in data inconsistencies.

Overall, the SMP, SMS, and MyUTHM function as isolated data silos, leading to data duplication and fragmented workflows. Without centralized and automated rule enforcement, the registration system operates as a basic data repository instead of a fully integrated academic management platform.

4.4 Comparison between Existing and Proposed Systems

This section compares the existing subject registration environment at UTHM with the proposed Subject Registration Enhancement Module. The comparison highlights limitations in the current system and demonstrates how the proposed subsystem improves usability and process efficiency.

Aspect	Existing System	Proposed System
System nature	Multiple separate systems (SMP, SMS, MyUTHM) operating independently	Integrated conceptual subsystem designed to align with UTHM digital ecosystem.
Registration Process	Students rely on separate tools for registration	Centralized registration portal with structured workflows
Timetable Validation	No real-time conflict checking and error detected after submission	Automated timetable conflict detection during selection
Data Consistency	Data duplication and inconsistencies across systems	Unified data model conceptually aligned with Enterprise Architecture across all related systems

Adminstrarive Workflow	High due to manual error correction and updation	Reduced through automated rules and approval
Mobility	Limited functional integration in mobile platforms	Designed to be responsive and usable across devices
User Experience	Complex navigation across multiple platforms	Simplified and centralize interface designed

Table 4.4 Comparison between Existing and Proposed Systems

4.5 System Requirements

The system requirements give the performance and features required of the Subject Registration Enhancement Module. They are divided into Functional Requirements, the behaviours which the system is supposed to behave, and Non-Functional Requirements, the quality traits of the system.

4.5.1 Functional Requirements

These are based on the requirements that are obtained as a result of the project scope and the sub-modules to be proposed.

i) Application Portal and Course Registration.

- The system will enable students to search a course list of the courses offered to their course of registration.
- The system will allow the students to find particular courses through keywords.
- The system will enable students to search courses according to the vacancy, schedule ability and credit hour restrictions.
- The system will enable students to choose and provide sections of the courses in order to register.

ii) Timetable Conflict Checker

- The system will cross-reference automatically the chosen courses with the courses that the student already takes and is intending to take.
- The system will identify time overlaps of course sections in real-time.
- The system will draw attention to the student before a submission is made on conflicted or duplicated subject selections.

iii) Credit Hour Validaton

- The system will compute the accumulated registered credit hours during that current semester.
- The system would check the aggregate credits with the definite minimum and maximum numbers of the programme of the student.
- The system will have warning signs in the case of choosing less than the minimum credits or more than the maximum credits.

iv) Add/Drop Approval Workflow

- The system will enable academic advisors to see a dashboard of add/drop requests among the students that are pending.
- The system will enable the advisors to either reject or accept the requests and add optional comments/remarks.
- A decision will be made automatically and therefore the system will send an automated email notification to the student.

v) Master data administrator course.

- The system will enable administrators to develop, edit, and store course master data.
- The system will enable the administrators to personalize the subjects of the sections and the quotas of capacity.
- The system will enable administrators to establish rules and prerequisites in registration of courses.

4.5.2. Non-functional requirements

These will guarantee the system is usable, reliable and integrated as part of the UTHM ecosystem.

i) Real-Time Performance: The system will carry out validation (timetable and credit hours) immediately when the selection process is being done so as to offer instant feedback to the user.

ii) Interoperability: The system will be based on Enterprise Architecture (EA) standards to facilitate the conceptual integration of the system with the current Student Information System (SMP) and Staff Information System (SMS) of UTHM.

iii) Mobility and Responsiveness: The system interface will be more than responsive and can be used in several devices (desktop, tablet, and mobile) to overcome the existing constraints in mobile integration.

iv) Data Integrity: This system will use a single data model to avoid data duplication and interdepartmental silos between the student and staff modules.

v) Security: To ensure safety of the system, all users (students, staff and admins) will be authenticated through the currently existing UTHM Single Sign-On (SSO) portal.

vi) Usability: The system will include a centralized dashboard that will make it easier to navigate the system, one will not have to change platforms and open numerous apps.

4.6 System Design

This section will be the technical specification of the Subject Registration Enhancement Module, and how the system will be configured to achieve the requirements stipulated. These design objects are used as a reference point in the implementation and integration of the system.

4.6.1 System Design – Enterprise Architecture

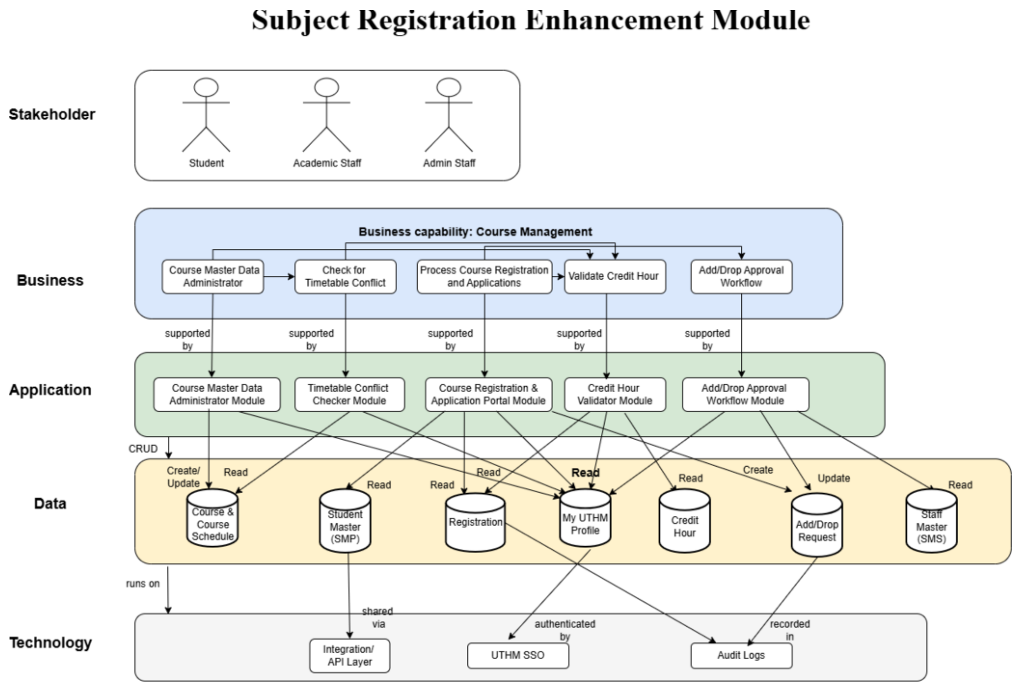


Figure 4.3 Enterprise Architecture (EA) for the Subject Registration Enhancement Module

The Enterprise Architecture (EA) for the Subject Registration Enhancement Module is designed to ensure seamless integration with the existing digital ecosystem of Universiti Tun Hussein Onn Malaysia (UTHM) while addressing the current issues of system fragmentation and manual workflows. By aligning the business, application, data, and technology layers, the architecture provides a holistic view of how the new module supports UTHM's digital transformation strategy.

4.6.1.1 Business Layer

The Business Layer defines the strategic capabilities required to improve academic management. The primary capabilities identified for this project include Subject

Registration Management, Course Management, and Academic Advisor Management. These are supported by specific business processes. First, is **Course Master Data Administrator** which oversees the configuration of course details, section quotas, and prerequisites. Secondly, **Check for Timetable Conflict** which proactively identifies scheduling clashes during the registration phase. Thirdly, **Process Course Registrations and Applications** which manages the core student enrolment workflow. Fourthly, **Validate Credit Hour** which enforces institutional rules regarding minimum and maximum credit limits. Lastly, **Add/Drop Approval Workflow** which digitises the interaction between students and advisors for course modifications.

4.6.1.2 Application Layer

The Application Layer translates defined business processes into interactive UI/UX components and functional prototypes developed using Figma. This layer ensures that registration logic is intelligent and cross-platform. The modules include **Course Master Data Administrator Module** which provides administrators with an interface to maintain and update program and course information, including quotas and prerequisites. Second, **Timetable Conflict Checker Module** executes real-time validation to detect scheduling overlaps during course selection. Third, **Course Registration & Application Portal Module** manages the core student enrollment workflow, allowing students to search, select, and submit course registrations. Fourth, **Credit Hour Validator Module** calculates and verifies semester totals to ensure compliance with institutional academic rules. Lastly, **Add/Drop Approval Workflow Module** digitizes the approval process, replacing paper-based or informal communication between students and advisors for course modifications.

4.6.1.3 Data Layer

The Data Layer defines the core data entities required to support integrated academic management. The primary entities identified for this project include Course & Course Schedule, Student Master (SMP) & Staff Master (SMS), Registration & Credit Hour, Add/Drop Request, and MyUTHM Profile. These are supported by specific data processes. First, **Course & Course Schedule** centralizes program and schedule information to ensure students access accurate, up-to-date details. Secondly, **Student Master (SMP) & Staff Master (SMS)** maintains comprehensive personal and academic profiles by interfacing with legacy systems. Thirdly, **Registration & Credit Hour** stores active enrollment records and validates semester credit totals. Fourthly, **Add/Drop Request** maintains a traceable record of course modification requests and advisor comments. Lastly, **MyUTHM Profile** synchronizes student-facing data to provide a cohesive experience across the mobile ecosystem.

4.6.1.4 Technology Layer

The Technology Layer defines the infrastructure and tools required to enable secure, real-time academic management. The primary components identified for this project include Integration / API Layer, UTHM SSO, and Audit Logs. These are supported by specific technology processes. First, **Integration / API Layer** uses REST APIs to facilitate real-time data exchange between the new module and SMP/SMS legacy systems. Secondly, **UTHM SSO** ensures secure and seamless user authentication through the existing Single Sign-On portal. Lastly, **Audit Logs** capture system activity to maintain accountability, traceability, and data integrity throughout the registration lifecycle.

4.6.2 System Architecture

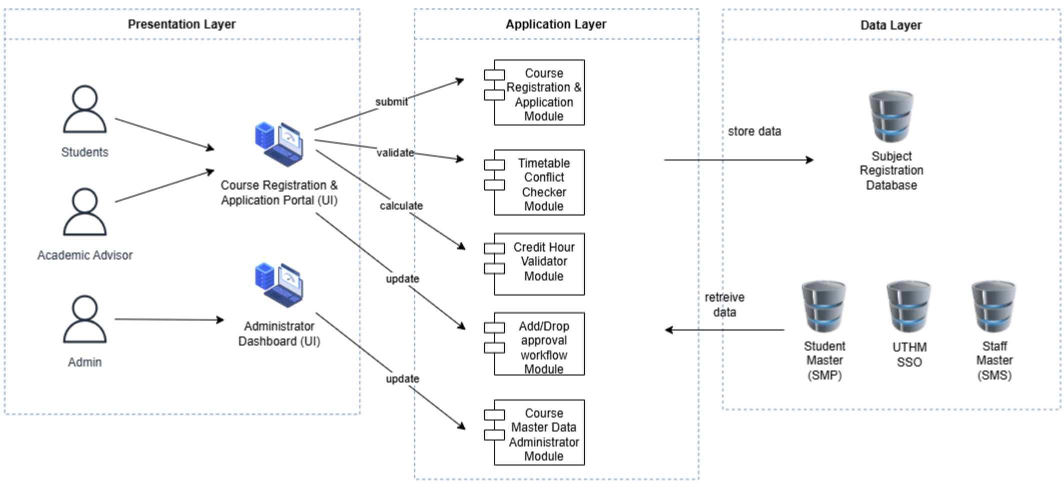


Figure 4.4: System Architecture diagram for Student Registration Enhancement Module

The diagram above shows the system architecture for the Student Registration Enhancement Module, it designed using a layered approach including Presentation Layer, Application Layer and Data Layer. This structure ensures a clear separation of concerns, allowing user interactions, business logic processing and data management to be handled independently for better maintainability and scalability.

In the presentation layer, it provides user interface for three different role such as students, academic advisors and administrators through the Course Registration and Application Portal and Administrator Dashboard. For the application layer, it having five core functional modules, timetable conflict checking, credit hour validation, add/drop approval workflow and administration. Lastly, the data layer store data in a core database named Subject Registration Database while integrating with existing UTHM master systems such as Student Information System (SMP) and Staff Information System (SMS) to retrieve authoritative data without duplication.

4.6.3 Use Case Diagram

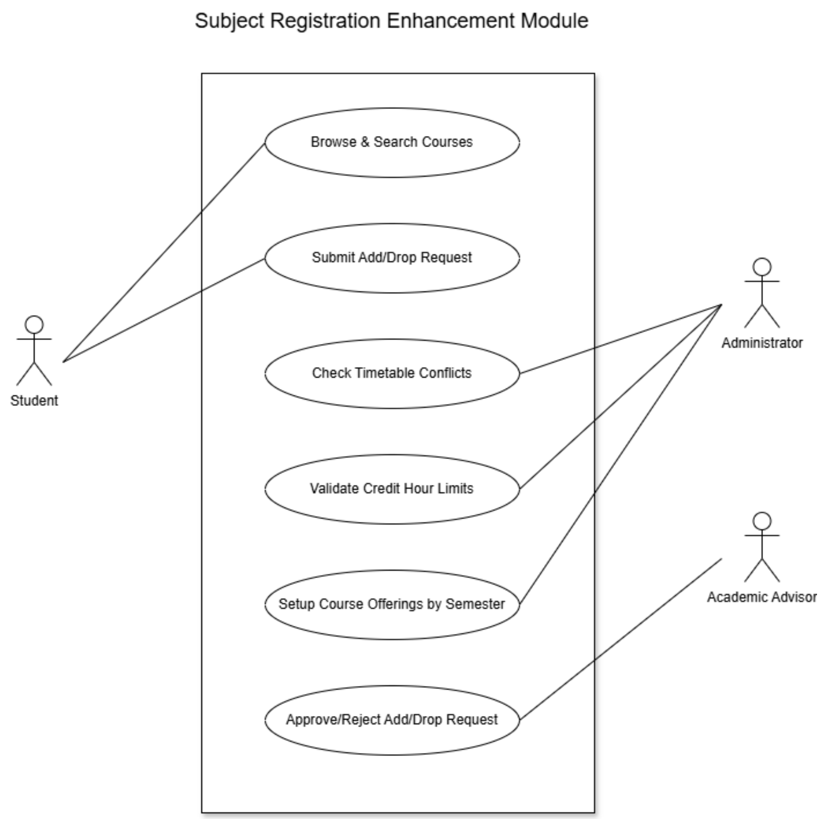


Figure 4.4 Use Case Diagram for the Subject Registration Enhancement Module

The Figure 4.4 Use Case Diagram represents the essential interactions of the Subject Registration Enhancement Module having three major roles: students, administrators, and academic advisors. The actions that students take are browsing and searching courses, verifying course timetable conflicts, verifying credit hour restrictions, and adding and dropping courses. The following use cases are indicative of the way the system aims at enhancing the accuracy of the registration and assisting the students in choosing subjects with confidence in the academic regulations.

The diagram also indicates that administrators define courses offered by semester along with subject lists and sections, and academic advisors revise and approve or deny add/drop requests with optional notes. This structure is significant to the report as it is straightforward on the user roles and system limits with the student's doing selection and validation, the administrators doing course data preparation and advisors doing academic approvals.

4.6.4 Database Design

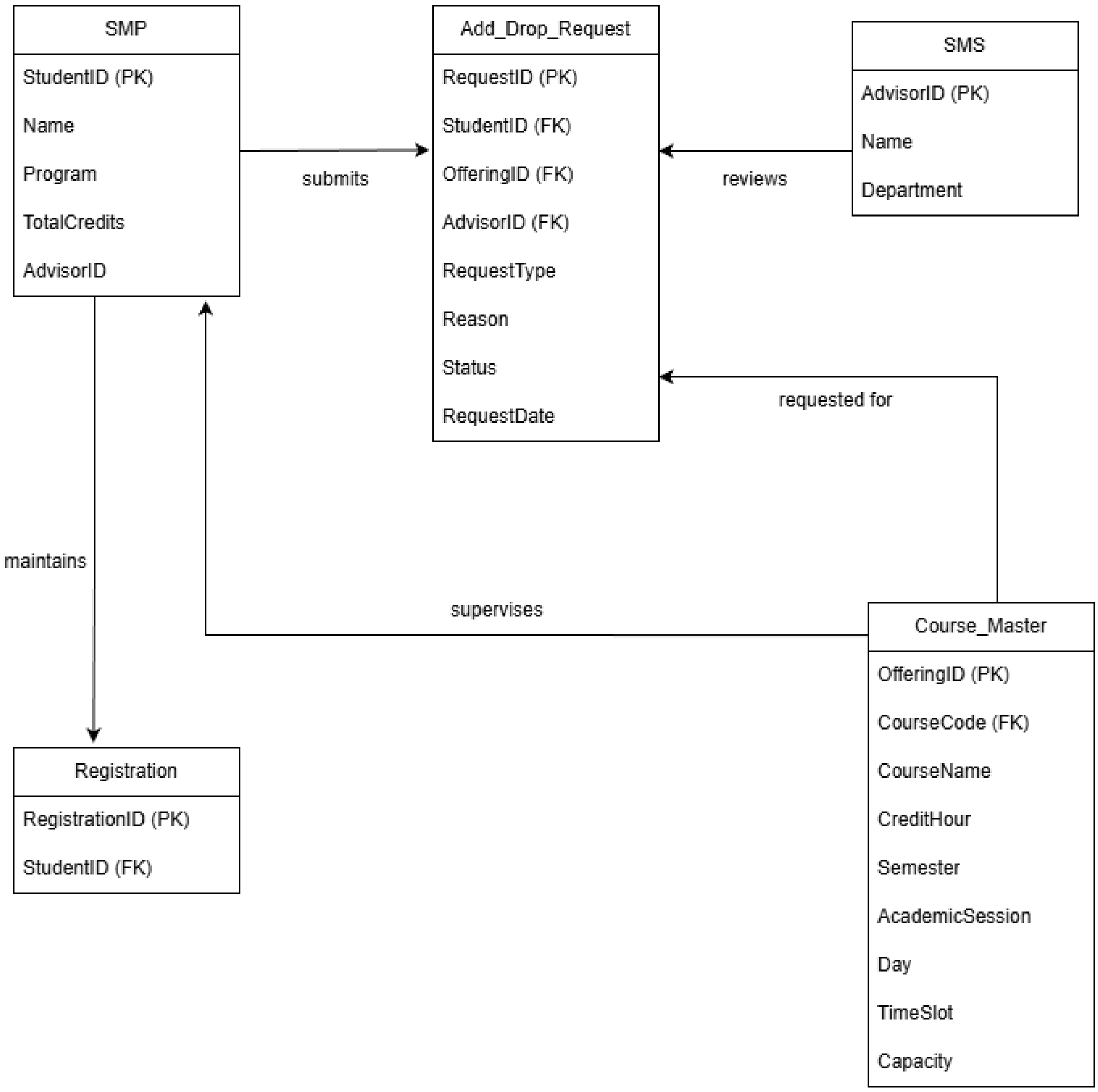


Figure 4.6.4 ERD Diagram for the Subject Registration Enhancement Module

The proposed system has an Entity-Relationship Diagram (ERD) written in Figure 4.5. This logical model of data represents the database design and identifies the database entities, attributes and relationships that are required to facilitate course registration, timetable validation and add/drop approval process.

4.6.4.1 Entities and Attributes

The system has six major entities that constitute the database holding the operational data of the system:

1. Student: This object includes the user profile. Matric_No is used as the Primary Key (PK), as it is used to distinguish every student. Examples of essential academic information that can be kept in it include Student_Name, Program_Code, and Total_Credit_Hours that are vital to the Credit Hour Validator module.

2. Course: This is where all subjects have their master data. Course_Code is the Primary Key (PK). It contains such attributes as Course_Name and Credit_Value that will be fetched during registration process in order to compute semester loads.

3. Section: This object represents a particular instance of classes based on a course. Section_ID is the Primary Key (PK). It will include schedule information Day, Time, and Venue. This is a fundamental element that the Timetable Conflict Checker relies on to determine the situation where there exist overlapping time slots.

4. Registration: It is an associative table which connects Student and Section. It captures the individual areas a learner has enrolled in. It contains a status attribute to follow the enrolment condition.

5. Add_Drop_Request: This object takes care of the change process. It connects a Student to a Section which they want to add or drop. Attributes are Request_Status and Advisor_Comment to give feedback.

6. Academic_Advisor: This is a person that represents the staff member, who handles approvals. The Primary Key (PK) is Staff_ID. It is also associated with the Add/Drop Request to create accountability of the decisions made.

4.6.4.2 Cardinality and Relationships

The following relationships are made by the diagram so that the data integrity is guaranteed:

- Course to Section: One Course may have one or more Sections. However, one Section may be in only one Course.

- Student to Registration: A Student may have numerous Registration records, constituting their semester schedule.

- Section to Registration: Section may be connected to many Registration records, which are the list of students attending that course.

- Student to Add_Drop_Request: A Student may make more than one Add_Drop_Request in a period of time.

- Academic_Advisor to Add_Drop_Request : An Academic_Advisor can provide various students with many different Add_Drop_Request that can be approved by the Academic_Advisor.

4.6.5 Sub-Module 1: Course Registration & Application Portal

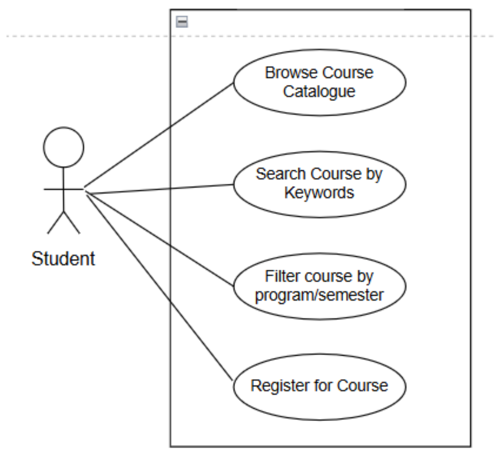


Figure 4.6.5.1: Use Case Diagram for Course Registration & Application Portal Module

The primary user of the module who interacts with the system to select and enroll in subjects is student. The use case “Browse Course Catalogue” allows the student to display and view the list of courses available specifically for their registered programme. The use case “Search Course by Keywords” enables students to find specific subjects within the repository. The use case “Filter Course by Program/Semester allows students to narrow down the course list based on vacancy status, timetable compatibility and credit hour limits. Lastly, the use case “Register for Course” is the final functional step where the student submits their subject selection to process the enrollment.

Use case: Course Registration and Application Portal Module
ID: UC01
Actor: Student
Preconditions: 1.Student is successfully authenticated through the UTHM Single Sign-On (SSO) portal 2. Student's academic profile and program details must be accessible from the Student Master (SMP) system. 3. The Course Master Data must be updated and available for the current semester.
Flow of events: 1. Student accesses the portal and the system displays a list of courses tailored to their specific registered program. 2. Student uses the Search and Filter functions to identify desired subjects. 3. Student selects the specific course sections they wish to enroll in. 4. The system temporarily holds the selection and performs a preliminary check against registration rules. 5. The Student clicks the "Register" button to submit their subject selection 6. The system processes the enrollment and generates a new registration record.
Postconditions: 1. A new Registration entity is successfully created and stored in the database. 2. The student's total registered credit hours are updated for further validation by the Credit Hour Validator.

3. The registered subjects are made available for the Timetable Conflict Checker to verify schedule integrity.

Table 4.6.5.1: Use Case Description for Course Registration and Application Portal Module

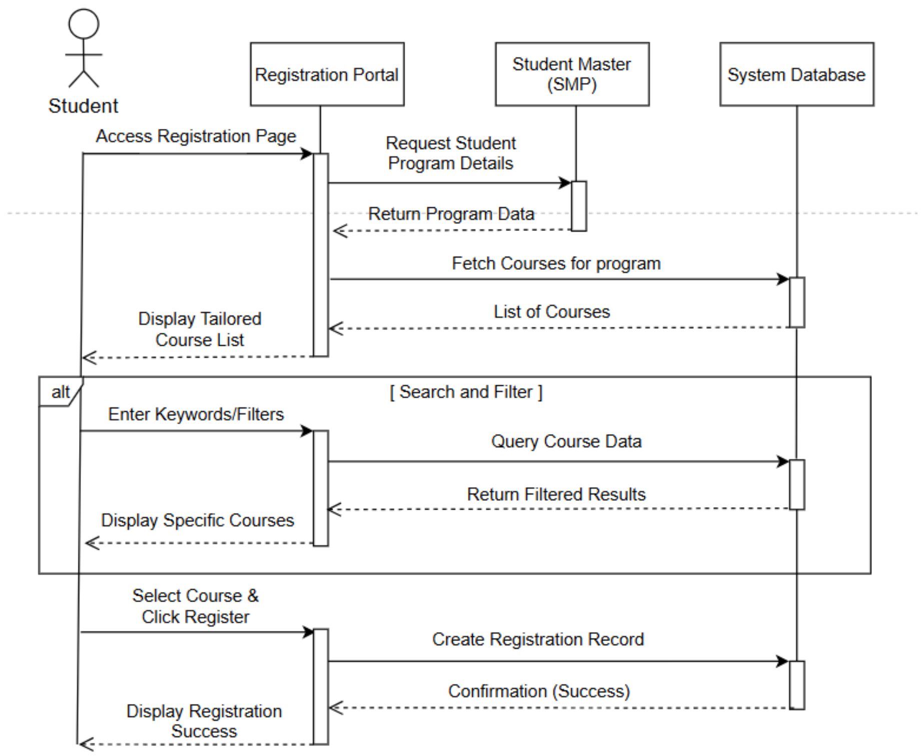


Figure 4.6.5.2 : Sequence Diagram for Course Registration & Application Portal

The sequence diagram for the Course Registration & Application Portal shows the systematic exchange of messages between the Student, Registration Portal, Student Master (SMP), and the System Database during the course registration lifecycle. The process starts when the Student accesses the registration page, prompting the Portal to request and receive specific program details from the Student Master (SMP) to ensure academic alignment. Simultaneously, the Portal fetches the relevant course list from the System Database and displays a tailored selection to the Student.

The workflow includes an alternative fragment for Search and Filter operations, where the Student can enter specific keywords or criteria. This action triggers a query to the System Database, which returns filtered results for the Portal to display back to the user. Finally, the Student completes the process by selecting a course and clicking the "Register" button, causing the Portal to send a command to the database to create a new registration record. Once the database confirms the successful creation of the record, the Portal displays a registration success message to the Student, concluding the transaction.

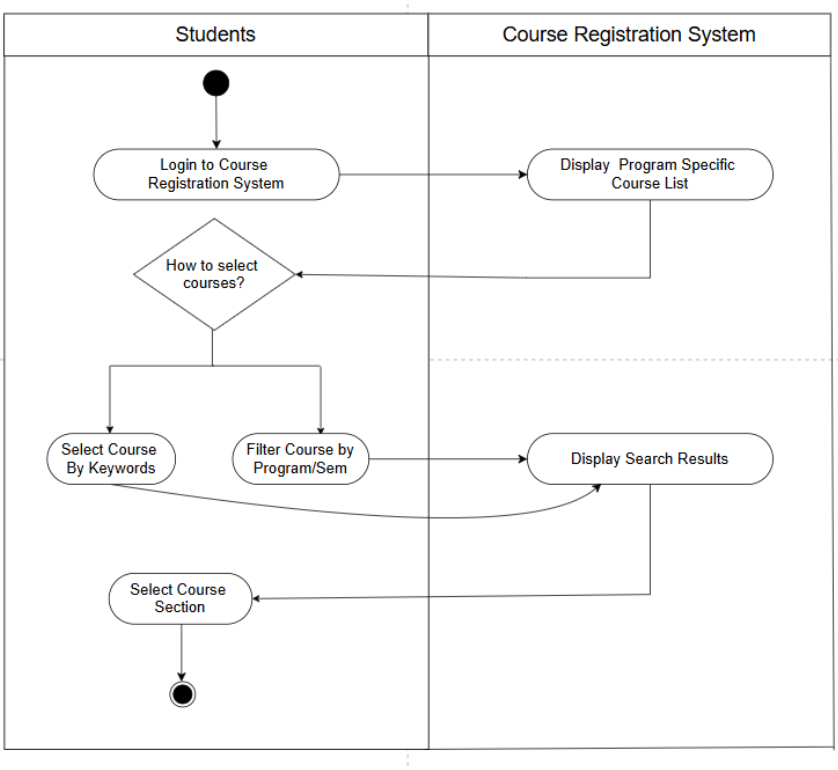


Figure 4.6.5.3 : Activity Diagram for Course Registration & Application Portal

The Activity Diagram for the Course Registration & Application Portal illustrates a collaborative workflow between the Student and the Course Registration System. The procedure begins in the Student swimlane when the user initiates a Login to the Course Registration System, which serves as the entry point for the academic module. Upon successful authentication, the system automatically executes the Display Program Specific Course List action, retrieving data from the Student Master (SMP) to present a tailored selection of subjects relevant to the student's specific degree.

Following this, the student encounters a decision point to determine how they will select their courses. This branches into three functional paths: the student may choose to Select Course By Keywords to find a specific known subject, Filter Course by Program/Sem to narrow down options based on vacancy or credit limits, or bypass these tools to browse the existing list directly. If a search or filter is applied, the flow transitions back to the system to Display Search Results, updating the student's view with the refined list of subjects. Once the desired subject is identified through any of these three paths, the student performs the final Select Course Section action to prepare the enrollment for processing, effectively completing the module's primary interaction cycle.

4.6.6 Sub-Module 2: Timetable Conflict Checker

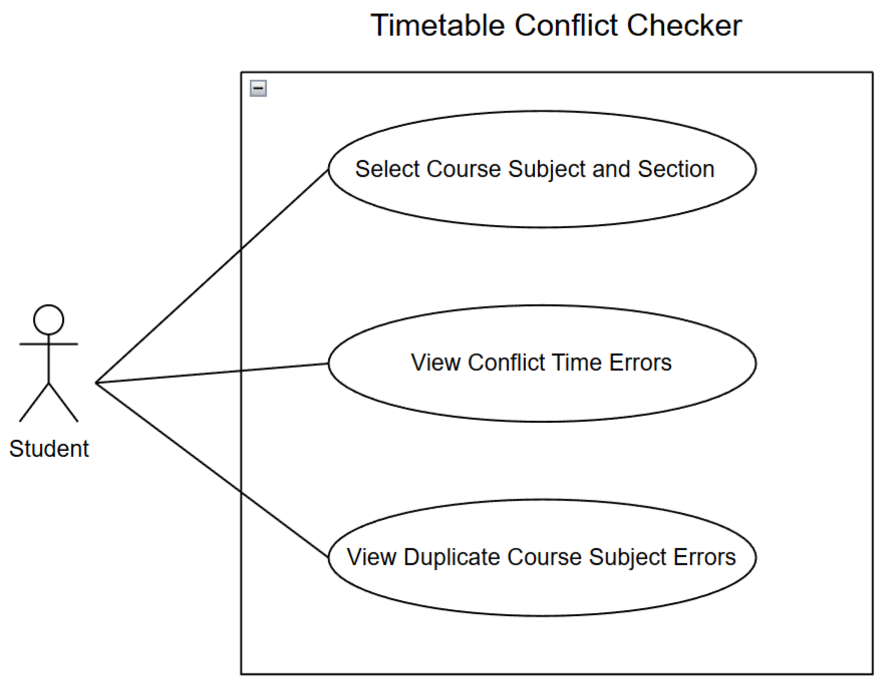


Figure 4.6.6.1 : Use Case Diagram for Timetable Conflict Checker Module

Figure 4.6.6.1 above shows the use case diagram for the Timetable Conflict Checker Module, where the system will check for any conflict between the registered course and section against the existing timetable. The use case "Select Course Subject and Section" allows student to register the chosen course subject and section. The use case

“View Conflict Time Error” checks for any conflict between the student’s registered course and section with the existing timetable. Lastly, the use case “View Duplicate Course Subject Errors” checks for any duplicated registration of the same course and section.

Use case: Timetable Conflict Checker
ID: UC02
Actor: Student
Preconditions: 1. The student can login through the UTHM Single Sign-On (SSO) system. 2. The student has access to the Course Registration and Application Portal. 3. The student is active and eligible to register for courses and sections.
Flow of events: 1.The student selects course subjects and their sections through the Course Registration Portal. 2. The system retrieves the schedule details for the selected courses and the student’s existing timetable. 3. The conflict checker will validate the chosen course and section against the existing timetable. 4. The student views the validation result. 4.1 If conflicts or duplicates are detected, then a message error will be displayed, and the student is not allowed to register. 4.2 If no conflicts or duplicates are detected, then a success message will be displayed and the Submit button will be enabled. 5. The student clicks the Submit button. 6. The system will update the student’s timetable. 7. The student can view and check the updated timetable.
Postconditions:

1. The system will allow the student to register a new course and section when no conflicts or duplicated data is found. 2. The student's timetable is updated in the system.

Table 4.6.5.1: Use Case Description for Timetable Conflict Checker Module

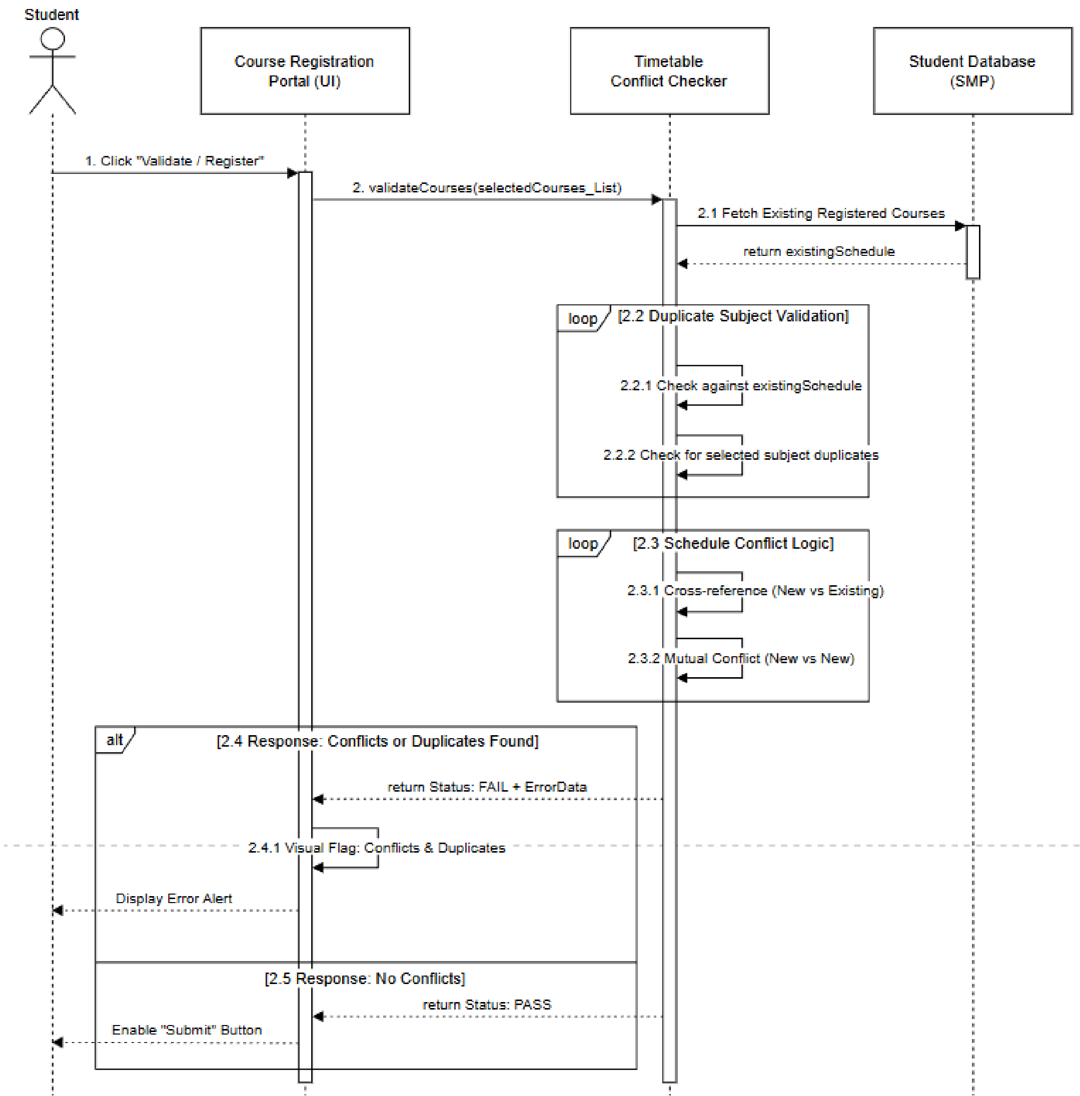


Figure 4.6.6.2 : Sequence Diagram for Timetable Conflict Checker Module

The sequence of interactions starts with the activation of the validation process by clicking the "Validate / Register" button on the Course Registration Portal (UI) by the

Student. This operation causes the interface to call the validateCourses operation and pass the selectedCourses_List to the Timetable Conflict Checker object. To be able to properly set a valid baseline of comparison, the checker module directly starts the process of synchronous data retrieval by making a query in the external Student Database (SMP). This step is important because it queries the current timetable of the student hence any new selections are compared with the latest academic records that are available in the legacy system of the university.

After the retrieval of the data, the system performs the main logic in two separate iterative loops. The first loop which is the Duplicate Subject Validation is used to verify the input that the student has not chosen the same subject twice. The second loop is the Schedule Conflict Logic which does a cross-reference of the time-slots of the new selections with the current schedule to identify any overlaps. The order ends with another fragment of the combination (alt) to process the results: in case of any contradictions or repetitions, the system gives an FAIL status and show an error message; otherwise the system gives a PASS status, which allows the finalisation of the registration as the "Submit" button is activated.

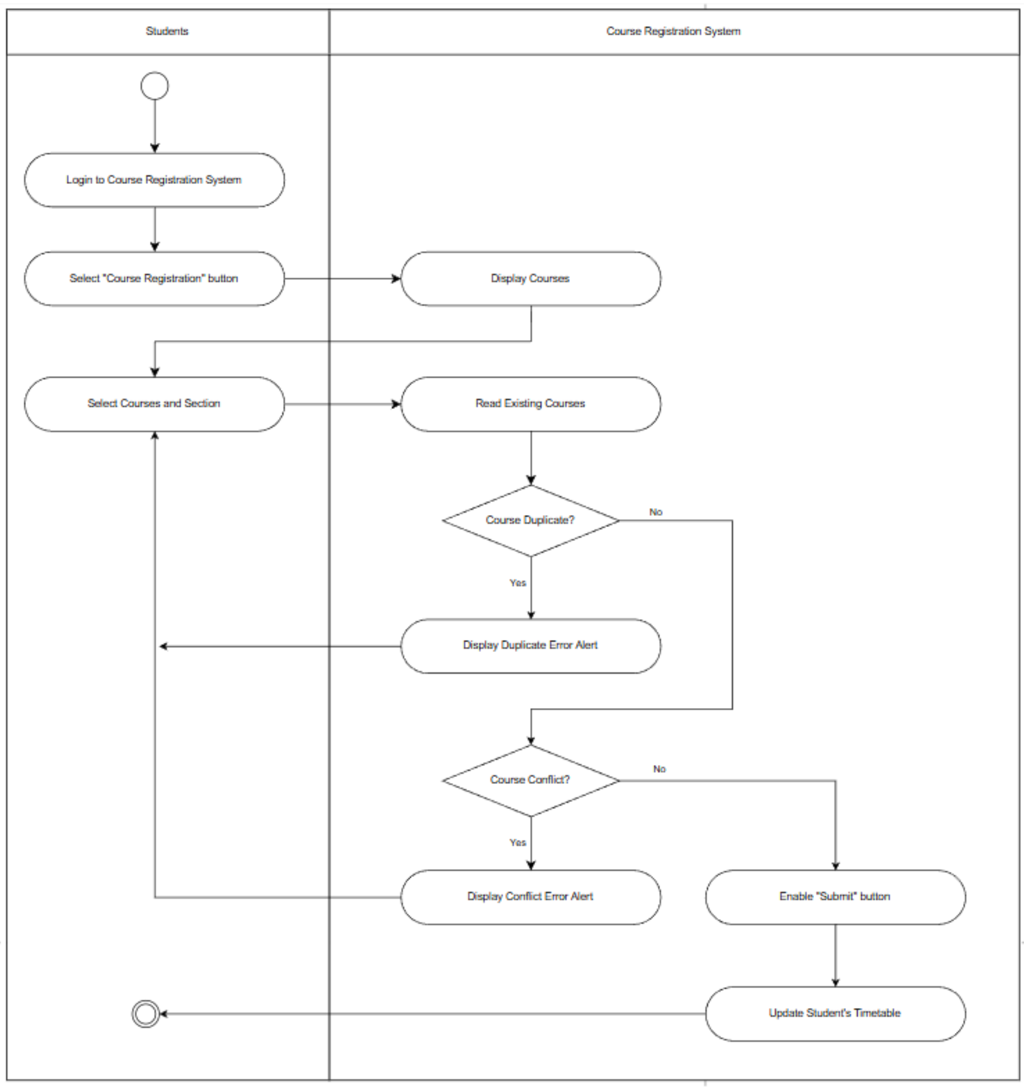


Figure 4.6.6.3 : Activity Diagram for Timetable Conflict Checker Module

The activity diagram shows the flow of interaction between the Student and the Course Registration System during the course registration process. The sequence starts when the student enters the system and presses the button of course registration. This act leads to the system showing the course catalogue available. The student would then choose the subjects and sections that they desired. This choice serves as a driving input that enables the system to access and read the already registered courses of the student as a form of setting a baseline of validation.

After retrieving the data, a sequential validation logic is followed by the system. The first one is that it checks whether the course chosen is a duplicate; if Yes, the system presents the Duplicate Error Message and returns to the stage of choosing the course.

In case of No, it goes to test timetable conflicts. Equally, when a time conflict is observed (Yes), a Conflict Error Message appears, and the student will be required to select again. The system will only enable the Submit button once there are no conflicts or duplicated course and section detected.

4.6.7 Sub-Module 3: Credit Hour Validator

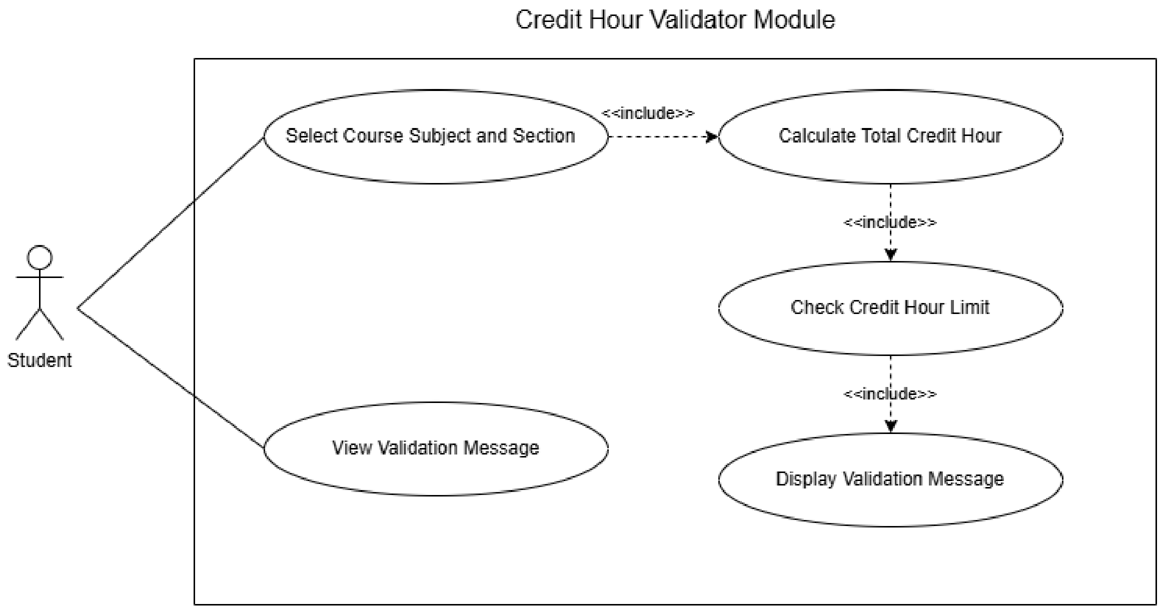


Figure 4. Use Case diagram for Credit Hour Validator Module

Figure 4. above shows the use case diagram for the Credit Hour Validator Module, which is about how students interact with the system to get respond on the credit hour validation process. First, students will select course subjects and their section. This module is automation where system will calculate the total credit hour by retrieving database values. After calculating, the system will check the credit hour limit whether students' credit hours reach minimum or over maximum. Validation messages will then pop out to show the results.

Use case: Credit Hour Validator Module
ID: UC03
Actor:

- Student
Preconditions: 1.The student can login through the UTHM Single Sign-On (SSO) system. 2.The student has access to the Course Registration and Application Portal. 3.Course and credit hour information exists in the Subject Registration Database.
Flow of events: 1.The student selects course subjects and their sections through the Course Registration Portal. 2. The system retrieves the credit hour value for each selected course from the database. 3. The system calculates the total registered credit hours automatically. 4. The system checks the calculated total against the minimum and maximum credit hour limits defined by the program. 5. The system displays a validation message indicating whether the selected credit hours are valid, below the minimum or exceed the maximum allowed limit. 6. The student views the validation result.
Alternative Flow: A1: Credit hours below minimum or exceed maximum 1. The system displays a warning message highlighting the credit hour issue. 2. The student may modify the selected courses and re-trigger the credit hour validation process.
Postconditions: 1.The student is informed of the credit hour validation result. 2. The system allows the student to proceed with registration if the credit hour requirement is valid or revise course selection if invalid.

Table 4.3 Use Case Description for Credit Hour Validator Module

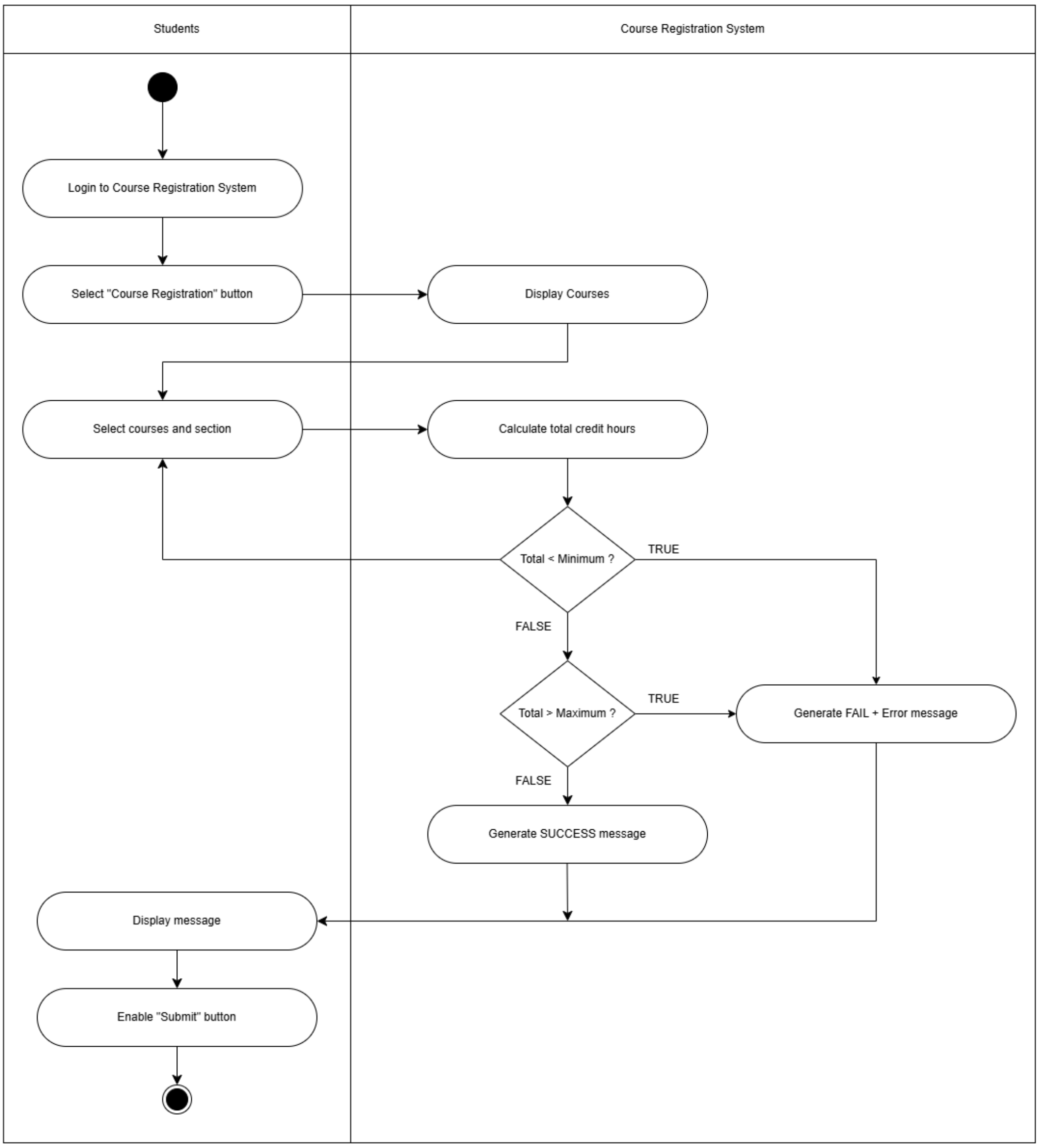


Figure 4. Activity diagram for Credit Hours Validator Module

The activity diagram of the Credit Hours Validator Module as illustrated in figure above depicts how the student and the Course Registration System interact during credit hour validation. This starts with the students registering into the system and choosing a course and then the system shows the available courses. The student then chooses course subjects and sections, and this will automatically cause the system to compute the total registered credit hours.

The system will compare the total credit hours with the minimum and maximum limit. In case the credit hours are less than the minimum or more than the minimum, an error message will be shown, otherwise a success message will be shown. When the system is successfully validated, the submit button is enabled and the student can register course.

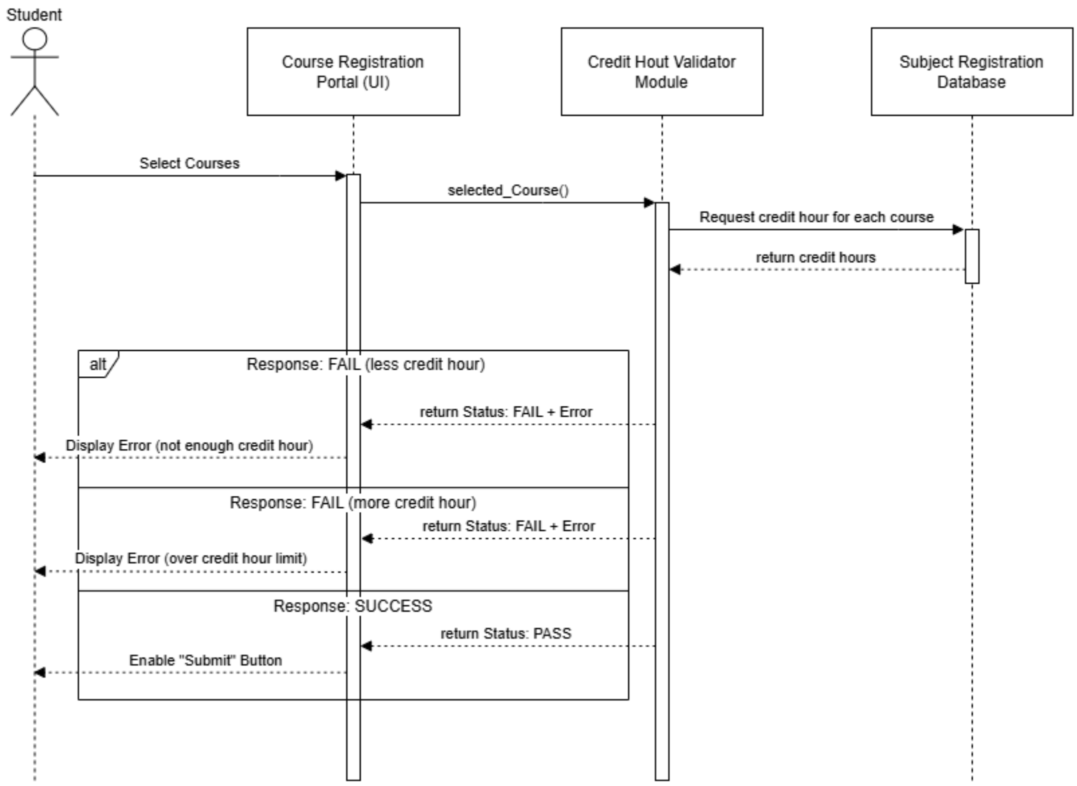


Figure 4. Sequence diagram for Credit Hours Validator Module

Figure above shows the sequence diagram of the Credit Hours Validator Module which shows the interaction between the student, Course Registration Portal, Credit Hour Validator Module and Subject Registration Database. This process starts when the student chooses courses through the registration portal which sends the information on the chosen course to the Credit Hour Validator Module. The validator then asks the database to provide him with credit hour data of each course he has chosen and is given the values.

According to the retrieved data, the system will assess the total credit hours and send validation outcome to the Course Registration Portal. In case the number of credit hours is less than the minimum or more than the maximum allowable number, an error message is presented to the student. On the other hand, when the credit hour requirement is satisfied, student to continue with course registration.

4.6.8 Sub-Module 4: Add/Drop Approval Workflow

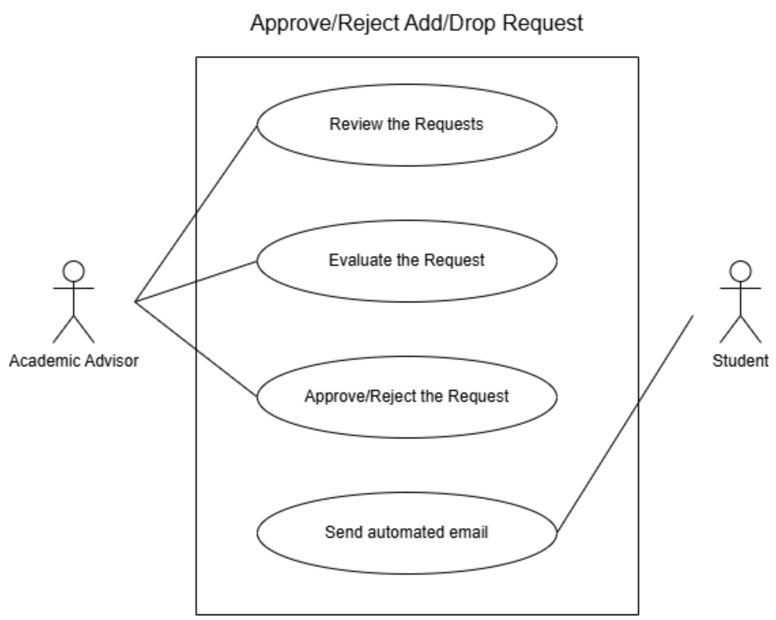


Figure 4. Use Case Diagram for Sub-Module Add/Drop Approval Workflow

In Figure 4. , the use case diagram describes the sub-module of add/drop approval workflow, which demonstrates the interaction between the students and the academic advisors in the process of course modification. The academic advisor in this workflow carries out the core decision-making activities, including its examination of received requests and assessment of each application according to academic rationale, schedule to schedule, and credit. Once the assessment is made, the advisor would then approve or reject the request and they can add brief remarks to the request to document and clarify.

It also contains a system triggered action of automated email notification that is triggered once the advisor completes their decision, and students are informed of the result of their submission. The module is important in the project because it ensures that there is a well-established and traceable approval process, less reliance on informal or manual communication, and efficient notification of students once a decision has been reached.

Use case: AddOrDropApprovalWorkflow
ID: UC04
Actor: - Academic Advisor - Student
Preconditions: 1.Student is authenticated and has submitted an add/drop request. 2.Request record exists in the system database. 3.Advisor has access permission to view student requests.
Flow of events: 1.Advisor opens the add/drop request dashboard. 2.Advisor reviews the submitted request details. 3.Advisor evaluates the request for academic justification and rule compliance. 4.Advisor approves or rejects the request. 5.System sends an automated email to the student with the decision.
Postconditions: 1.Request status is updated (approved/rejected). 2.Advisor decision is stored for record. 3.Student receives notification via email.

Table 4.4 Use Case Description for Add/Drop Approval Workflow

The table above describes how to do it when a student requests to change subjects. The request is sent by the student, and the academic advisor examines the information and passes or denies it according to the academic regulations. Once the decision has been made, the system changes the request status and provides an automated email to the student. The decision made by the advisor is captured in form of a record making the process transparent, responsible and effective.

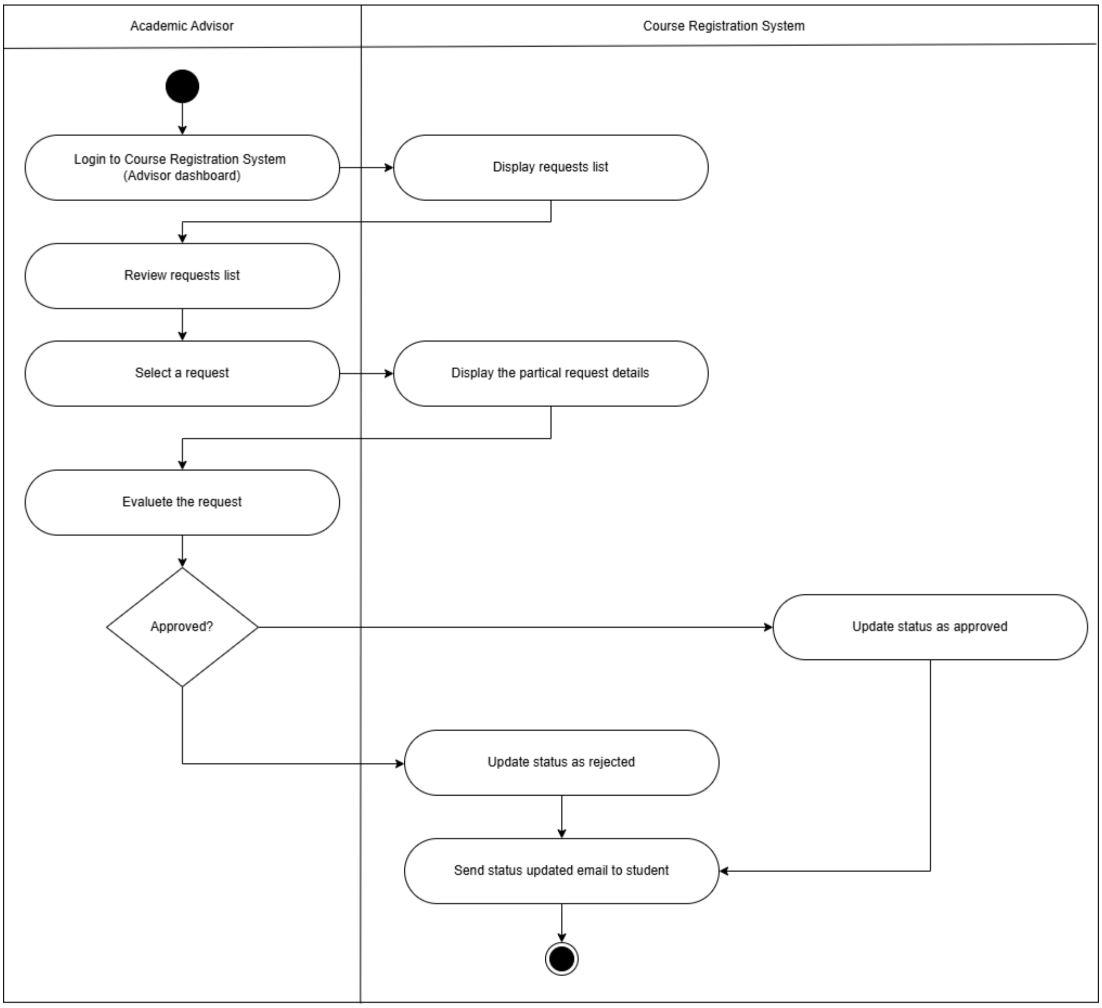


Figure 4. Activity Diagram for Add/Drop Approval Workflow

The activity diagram defines movement of operations in the Add/Drop Approval Workflow. The process begins with the academic advisor logging into the Course Registration System whereby the system gives him/her a list of requests submitted to the system through add/drop requests. The advisor picks a request and reviews it in terms of academic compliance. This workflow further diverges to a decision point, in the case where the request was denied, the status is imparted to denied. After a decision has been made, the system will send an automated email notification to a student. The notification being sent after the activity is ended represents a clear and sequential and traceable process.

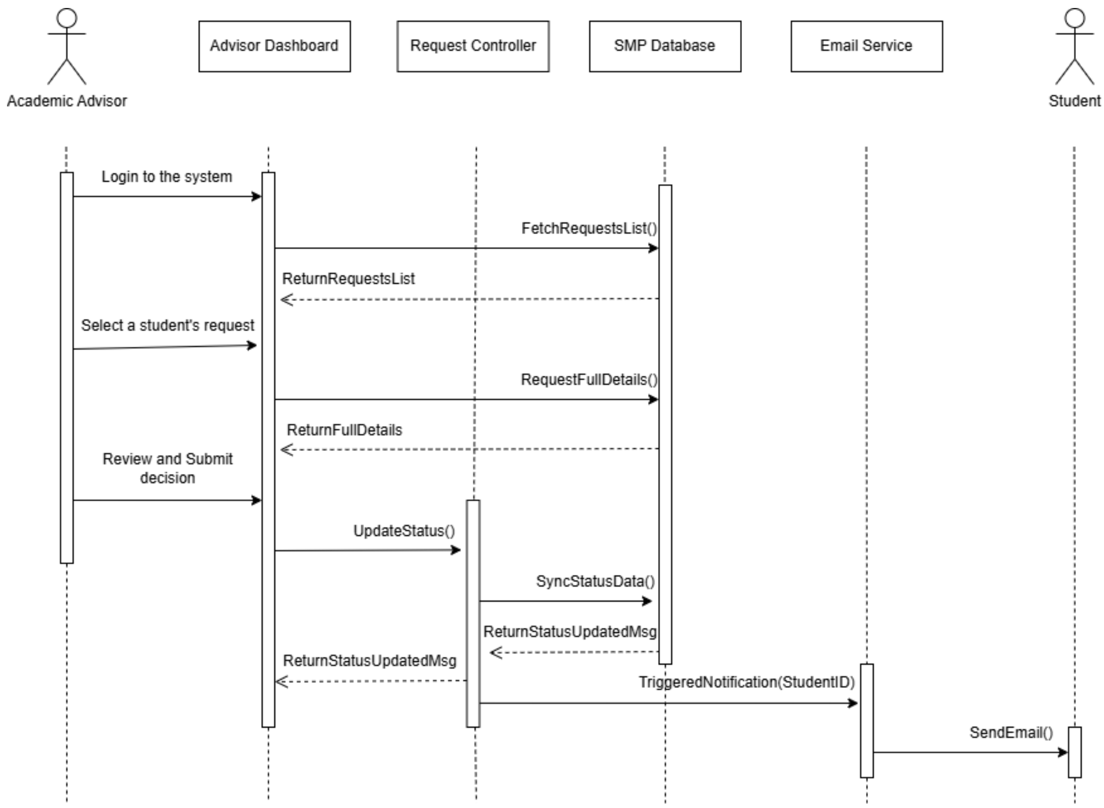


Figure 4. Sequence Diagram for Add/Drop Approval Workflow

The sequence diagram displays the message exchange of the Add/Drop Approval Workflow. The proves starts when the academic advisor enters via advisor dashboard in which the system will retrieve the list of add/drop requests stored in SMP Database. Once the advisor has chosen a particular student request, the system will fetch and display the complete request details that can be viewed. After the decision is submitted by the advisor, the system changes the request status and renews the updated data in the database. The system then activates the Email Services to send alarm to the student containing the final result of the approval. The figure shows the organized backend interaction that guarantees updates to data and automatic student notification.

4.6.9 Sub-Module 5: Course Master Data Administrator

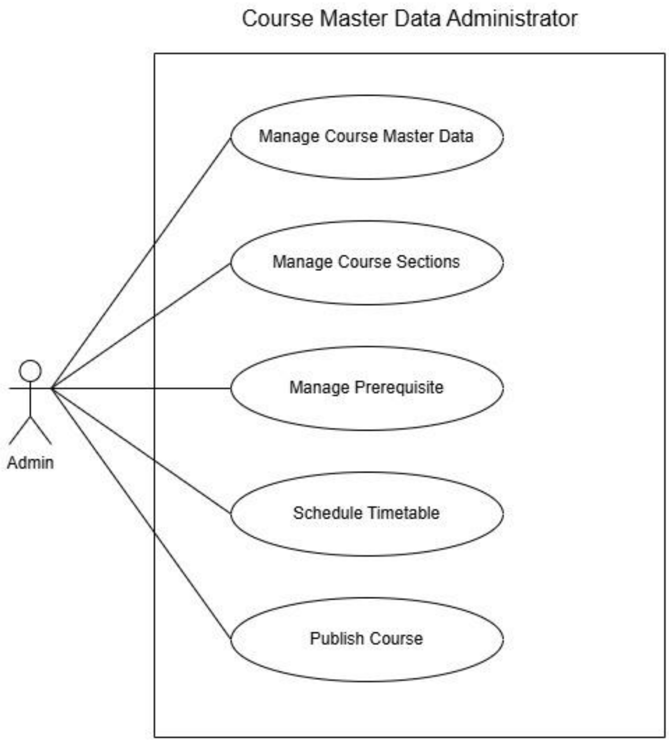


Figure 4. Use Case Diagram for Sub-Module Course Master Data Administrator

In Figure 4. , the use case diagram illustrates the Course Master Data Administrator sub-module, which focuses on administrative control over course-related information within the Subject Registration Enhancement Module. This sub-module demonstrates the interaction between the administrator and the system in managing academic course offerings before they are made available to students.

The administrator is responsible for performing core management activities such as creating new courses, updating existing course information, managing course sections, defining prerequisite rules, and scheduling timetables. These actions ensure that course data is accurate, consistent, and compliant with academic regulations.

The system supports the administrator by performing automated validations, including timetable conflict checks and prerequisite consistency checks, before allowing course data to be saved and published. This sub-module is essential as it establishes a structured and centralized approach to course data management, reduces manual errors, and ensures reliable course information across the registration system.

Use case: Course Master Data Administrator
ID: UC05
Actor: - Administrator
Preconditions: 1. Administrator is authenticated and authorized to access the admin dashboard. 2. Course Management Function is available in the system. 3. Basic academic master data exists in the system.
Flow of events: 1. The administrator logs into the system and accesses the Admin Dashboard. 2. The administrator selects the Course Management function. 3. The administrator creates or updates course information. 4. The administrator manages course sections, prerequisites, and timetable scheduling. 5. System validates course data for conflicts and rule consistency. 6. The administrator saves and publishes the course information.
Postconditions: 1.Course and section data are stored in the system database. 2. Validated course information is published and made available for registration.

Table 4.4 Use Case Description for Add/Drop Approval Workflow

The table above describes the process of managing course master data within the system. The administrator performs administrative tasks related to course creation and maintenance, while the system ensures data accuracy through validation mechanisms. Once validated, the course data is saved and published, allowing it to be used reliably in the subject registration process.

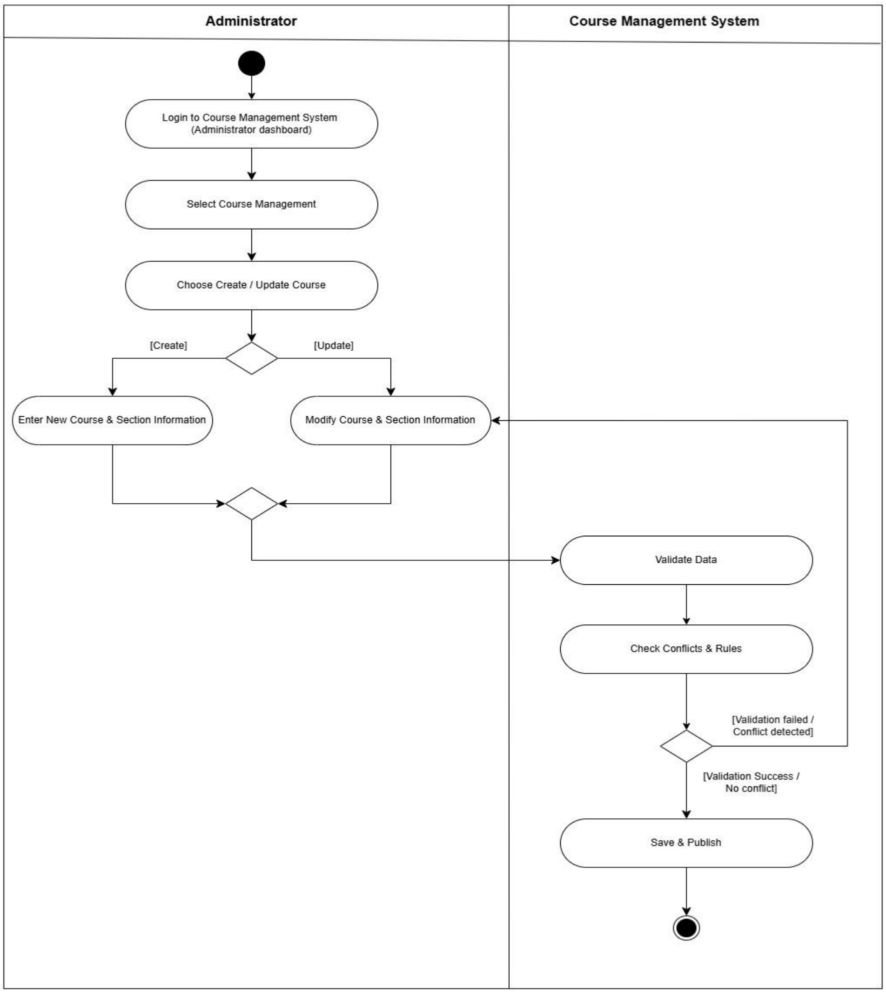


Figure 4. Activity Diagram for Sub-Module Course Master Data Administrator

The activity diagram illustrates the operational flow of the Course Master Data Administrator sub-module. The process begins when the administrator logs into the system and accesses the Course Management function through the admin dashboard. The administrator then performs actions such as creating or updating course information, managing course sections, defining prerequisite rules, and scheduling timetables.

The workflow proceeds to a validation stage where the system checks for timetable conflicts and prerequisite consistency in real-time. If validation fails, the administrator is redirected back to the modify page to revise the input. If the validation is successful, the course data is saved and published. The activity flow ensures that only validated and non-conflict course information is made available in the system.

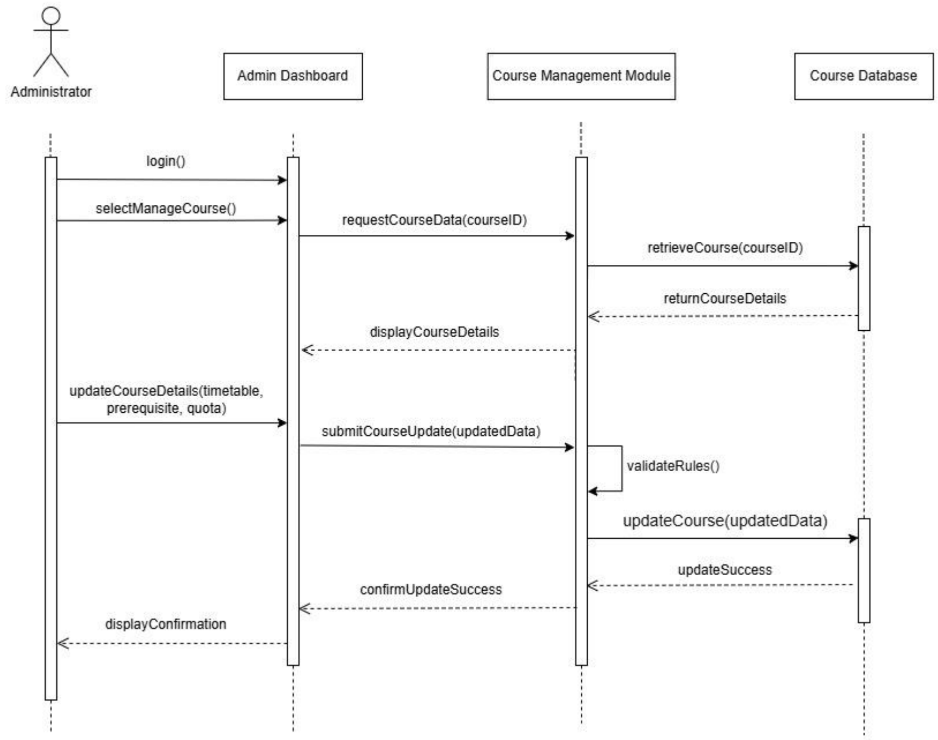


Figure 4. Sequence Diagram for Sub-Module Course Master Data Administrator

The sequence diagram presents the interaction between the administrator, the admin dashboard, the course management module, and the course database. The process starts when the administrator submits a request to create or manage course information via the admin dashboard. The request is processed by the course management module, which performs validation checks before interacting with the database.

Upon successful validation, the updated course data is stored in the database, and a confirmation is returned to the administrator. The sequence diagram demonstrates a structured interaction flow that ensures data integrity and centralized control of course master data within the system.

4.7 Summary

CHAPTER 5

CONCLUSION

5.1 Introduction

This chapter concludes the Subject Registration Enhancement Module project by summarizing the overall system development and its contribution to improving subject registration at Universiti Tun Hussein Onn Malaysia (UTHM).

This chapter also highlights the system's achievements, discusses limitations encountered during development, and proposes future improvements. Overall, it reflects on how the project supports UTHM's digital transformation and Smart Campus initiatives through the application of Enterprise Architecture principles.

5.2 System Contribution / Achievements

5.3 System Constraints

The proposed Subject Registration Enhancement Module is developed as a high-fidelity prototype using Figma to simulate user interactions, workflows, and validation logic. As a result, the project does not include real backend implementation, live databases, or deployment within UTHM's production environment.

Since UTHM currently does not operate an integrated enterprise system, this project represents a proposed Enterprise Architecture-based solution, where different submodules are designed separately by project members. Existing platforms such as SMP, SMS, and MyUTHM are treated as conceptual reference systems, and all integrations are illustrated through architecture diagrams and mock data rather than the actual system access.

The project scope is further constrained to academic subject registration functions, particularly course registration, real-time timetable checking, credit hour validation, add/drop approval, and course master data administration. Other academic modules such as financial billing, examination scheduling, graduation clearance, and student performance analytics are either handled by other project groups or not considered within the scope of the whole system.

Due to academic time constraints, system evaluation is limited to design-level validation using diagrams and prototype walkthroughs, without performance testing, real-user deployment, or system integration. Thus, the system's feasibility is only assessed at the conceptual and design level within the context of this project.

5.4 Future Suggestions

5.5 Summary